



Published in final edited form as:

Emotion. 2020 August ; 20(5): 750–760. doi:10.1037/emo0000590.

Creating Emotional False Recollections: Perceptual Recombination and Conceptual Fluency Mechanisms

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Abstract

We investigated the impact of two hypothetical mechanisms of episodic memory reconstruction—perceptual recombination and conceptual fluency—on objectively measured recollection accuracy and false recollections of neutral and emotional stimuli. Participants encoded negative, neutral, and positive pictures depicting objects and scenes (i.e., target pictures), each accompanied with a descriptive verbal label (e.g., “boy crying at funeral”, “wooden basket on floor,” “four chimpanzees laughing together”). Next, they encoded fragmented pictures of some of the scenes they did and did not earlier see (perceptual misinformation), or they received multiple presentations of the corresponding verbal labels (conceptual misinformation). Recollection of target pictures was then tested, using labels as retrieval cues. We had three key findings in each of two experiments. First, as in our prior work, both perceptual and conceptual misinformation significantly increased false recollection judgments of nonstudied pictures, including high confidence errors. These effects implicate perceptual recombination and conceptual fluency mechanisms. Second, these misinformation effects generalized across all emotional categories, implicating separable roles of these two mechanisms on emotional recollections. Finally, conceptual misinformation was less likely to influence negative than neutral recollection errors, providing new evidence that emotion can improve retrieval monitoring accuracy and reduce false memories based on conceptual fluency (i.e., an emotional distinctiveness heuristic).

Keywords

emotional memory; false memory; false recollection; perceptual recombination; conceptual fluency

Introduction

The recollection of emotional events is often accompanied with a subjective sense of vividness but research indicates that emotional events are susceptible to false recollection, or the subjective experience of conscious recalling some detail or event that did not actually happen. One complexity of this research literature is that emotional events have sometimes been found to be more prone to false recollection and other times less prone for reasons that

are not yet fully understood. For example, on episodic memory tests for words or pictures, a common finding is greater false recognition of nonstudied emotional than neutral items (e.g., Bessette-Symons, 2018; Bookbinder & Brainerd, 2017; Brainerd et al., 2008; Gallo, Foster, & Johnson, 2009; Gong, Xiao, & Wang, 2016; White, et al., 2014). Such findings have been attributed to the propensity for emotional events to share conceptual or semantic features or otherwise share an “emotional gist”. As a result, emotional studied events are more likely to activate nonstudied emotional events relative to neutral events, increasing false recognition for emotional events.

In contrast to these findings, it also has been found that emotional events can be less prone to memory distortion than neutral events, at least for negative emotional content (e.g., Farris & Toggia, 2018; Kensinger et al., 2007; Kensinger & Schacter, 2006; Pesta, Murphy, & Sanders, 2001; Zheng et al., 2018). According to Schacter, Gallo, and Kensinger (2007), people often expect emotional events to be recollected more vividly, and expecting vivid or perceptually detailed recollections might help reduce false recognition of nonstudied events that only feel familiar or vaguely recollected (i.e., an emotional distinctiveness heuristic). Although some studies relevant to this issue have conflated emotional and conceptual or semantic distinctiveness (Pesta et al., 2001), Choi, Kensinger, and Rajaram (2013) avoided these issues by testing recognition for emotional pictures and controlling for conceptual relatedness across emotional categories, and they found that emotional items were equally or less likely to be falsely recognized as emotionally neutral items on a recognition memory test. This study, therefore, provided stronger support for the emotional distinctiveness hypothesis, although the effect was not obtained in all of the experiments.

As articulated by Bookbinder and Brainerd (2016), emotion can simultaneously impact multiple (and sometimes competing) cognitive processes, and emotion itself has different facets, thereby making generalizations across areas of emotional memory research difficult. For example, the aforementioned research indicates that emotional events can be more prone to gist-based influences that can drive false remembering, but these events also may trigger heightened memorial expectations that might reduce false remembering. Because of this complexity, researchers have started to move away from asking questions like “are emotional events more or less susceptible to memory distortion?” and, instead, they are now asking how the multiple processes that contribute to memory accuracy and distortion might differentially contribute to emotional and neutral events. In this spirit, the current experiments aimed to investigate the extent that two hypothetical mechanisms of memory reconstruction might differentially contribute to the recollection of emotional and neutral events.

Two Reconstructive Mechanisms

Doss, Bluestone, and Gallo (2016) proposed two hypothetical mechanisms of memory construction that occur during retrieval. The first mechanism is perceptual recombination. According to this hypothesis, the erroneous perceptual content that characterizes false recollections can originate from previously experienced events that are stored in episodic memory. These perceptual details can be fragmented or dissociated from their instantiating events in memory, providing fuel for reconstructive processes that cause false memories.

The idea that different kinds of features or details can be erroneously confused across different events in memory has been proposed by many researchers (e.g., Dodson, Bawa, & Slotnick, 2007; Gallo & Roediger, 2003; Lampinen, Neuschatz, & Payne, 1999; Lyle & Johnson, 2006), although the perceptual recombination hypothesis focuses exclusively on perceptual features from prior experience to differentiate it from a conceptual fluency mechanism.

The second mechanism is conceptual fluency. According to this hypothesis, the erroneous perceptual content of false recollections originates from mental imagery, as opposed to actually perceived events. People might mentally imagine events and then later falsely retrieve them, or they might mentally imagine events as part of the retrieval process itself, as when someone uses mental imagery to cue or trigger their memories and monitor them for accuracy. In either case, if these mentally imagined perceptual details are generated in a way that makes them seem familiar or fluent, potentially due to increased semantic activation or conceptual priming, then they also might provide the fuel for a reconstructive process that causes false memories. The idea that conceptual fluency can be misattributed to prior experience has been proposed by many researchers (e.g., Gallo & Roediger, 2003; Kurilla & Westerman, 2008; Rajaram & Geraci, 2000; Whittlesea, 2002), although these ideas sometimes focus on familiarity-based false recognition as opposed to perceptually-detailed false recollection.

Doss et al. (2016) reported the first set of experiments that were specifically designed to isolate perceptual recombination and conceptual fluency effects on the false recollection of nonstudied pictures. To do this, Doss et al. developed a multi-phase misinformation task that involved the presentation of two kinds of misinformation: perceptual and conceptual. In the first phase (encoding phase), participants studied verbal labels corresponding to pictures of objects (e.g., “wine,” “puppy,” “lemon”) with half of these labels followed by the corresponding picture. These items would later serve as targets on the recollection test, whereas on the other half of trials, the verbal label was presented alone, later to serve as studied lures. In the next phase (misinformation phase), stimuli were presented to distort memory, though from the participants’ perspective, this was just additional information. There were two kinds of misinformation: perceptual misinformation (nonstudied pictures that had been scrambled) and conceptual misinformation (nonstudied verbal labels). Finally, an episodic memory test was given (cued recollection phase). On this test, participants were presented with all of the previously studied verbal labels as memory cues, and they were asked to press “yes” only if they recollected having studied an associated target picture during the encoding phase.

Doss et al. (2016) reported three key effects for present purposes. First, false alarms were greater for lures associated with conceptual misinformation compared to a no misinformation foil baseline, showing that the activation of the concept at the semantic level (without picture features) was sufficient to increase false alarms. Second, false alarms on the recollection test were greater for lures associated with perceptual than conceptual misinformation, even though each kind of lure had been earlier studied and hence was highly familiar. Doss et al. argued that the presentation of fragmented pictures fueled a perceptual recombination process that, in turn, increased false alarms above the conceptual

misinformation baseline. Finally, false alarms to misinformation lures were made with medium confidence on average (e.g., 2 out of 3), despite instructions to only respond “yes” when recollecting an intact target picture from the original encoding phase. These findings suggest that a false recollection process had influenced participants’ false alarms, driving high confidence errors in some cases, in addition to any potential effects of a more vague sense of guessing based on familiarity alone, which might elicit relatively low levels of confidence. (Of course, false recollections may be made with either high or low confidence, our point here is that by comparison, familiarity-based guesses on a picture recollection test should be made with lower confidence.)

The Current Experiments

The main goal of the current experiments was to investigate the impact of perceptual recombination and conceptual fluency on neutral and emotional recollections. Doss et al. (2016) found significant effects of perceptual and conceptual misinformation on emotionally neutral recollections, and here we aimed to replicate and extend these effects across a broader range of picture stimuli, including emotionally positive and negative as well as neutral pictures of common objects and scenes. Such replication and extension would further establish the robust and reliable nature of the empirical effects reported by Doss et al. and provide a test for our specific hypotheses. Based on the emotional memory literature, we had two contrasting (but not mutually inconsistent) hypotheses: the emotional enhancement hypothesis and the emotional distinctiveness heuristic hypothesis.

According to the *emotional enhancement hypothesis*, the same processes that tend to improve memory encoding for emotional over neutral items (e.g., enhanced attention, elaboration, and physiological arousal) also would benefit the encoding of misinformation, making this misinformation more accessible at retrieval and hence, more likely to drive false recollections. All other factors being equal, this hypothesis predicted greater perceptual recombination and conceptual fluency effects on false alarms for the emotional relative to neutral items.

According to the *emotional distinctiveness heuristic hypothesis*, people are more likely to expect perceptually detailed recollections for emotionally arousing events compared to neutral ones, thereby helping them avoid recollection errors to new events that are only familiar or vaguely recollected. This hypothesis predicted reduced perceptual recombination and conceptual fluency effects on false alarms for emotional items relative to neutral items, again all other factors being equal. Alternatively, such an emotional distinctiveness heuristic might help people monitor all sources of memory distortion (e.g., misinformation for studied lures and idiosyncratic item associations for nonstudied foils), thereby reducing false alarms to all emotional lures regardless of the misinformation condition. In each case, this hypothesis claims that emotion will increase a retrieval monitoring process to reduce false recollections, in contrast to the emotional enhancement hypothesis.

To investigate these hypotheses we conducted two experiments, reported here as Experiments 1 and 2. Each of these two experiments used the same modified version of the Doss et al. (2016) procedures, described below in the context of Experiment 1. By design,

the only difference between Experiments 1 and 2 reported here was in the timing of the 24-hour delay period, which allowed us to investigate the effects of delay on the reconstructive mechanisms of episodic memory. Doss et al. found a significant effect of perceptual and conceptual misinformation on an immediate memory test, but there was no effect of conceptual misinformation when a 24-hour delay was introduced between the misinformation and testing, indicating that conceptual fluency effects fade more rapidly over time than perceptual recombination effects. We sought to further explore these timing effects across Experiments 1 and 2 and evaluate how they might interact with emotional memory effects which tend to be more prominent after a delay (Yonelinas & Ritchey, 2015).

Experiment 1

Experiment 1 used a modification of the Doss et al. (2016) procedure to investigate the susceptibility of emotionally negative, neutral, and positive recollections to perceptual recombination and conceptual fluency. To do this, we constructed a new set of neutral and emotional picture stimuli, and we developed a four-phase misinformation task.

With respect to the stimuli, our goal with these experiments was to investigate the impact of emotion on recollection accuracy and distortion, independent from the semantic or conceptual relatedness shared between the pictures and concepts represented in the set of stimuli. Prior work has shown that emotional items can be more conceptually related than neutral items (e.g., Talmi & Moscovitch, 2004). To the extent that this factor can drive false recollection differences between emotional categories (see Gallo et al., 2009), it would represent a confound in the current procedure. To avoid this potential confound, the labels developed for the current experiments were created so that the same kinds of concepts (e.g., people, animals, objects, scenes) would be represented with similar frequency in each of the three emotional valence categories (negative, neutral, positive). Moreover, we selected labels and stimuli to be conceptually distinct from one another, so that memory errors were unlikely to be attributed to participants making label confusion errors between studied and nonstudied stimuli.

With respect to the task, there were four phases. The first was the picture encoding phase. Participants were presented with emotional and neutral scenes and their verbal labels (these would later be targets) intermixed with verbal labels without pictures (these would later be lures). The second phase was a perceptual misinformation list, which presented verbal labels and scrambled versions of the corresponding picture. The third phase was the conceptual misinformation list, which presented verbal labels without any other perceptual information. In this way, the familiarity of items presented in the two misinformation lists was quite high, requiring recollection to accurately discriminate between them. Importantly, as in Doss et al. (2016), misinformation was presented for both targets and lures to preclude a recall-to-reject strategy. The final phase was the picture recollection test, which presented labels from the first list, as well as brand new labels (nonstudied foils), as possible retrieval cues. On this test, participants were instructed to say “yes” if they recollected seeing an intact picture of the scene from the first list, regardless of what they may remember from a different list and then to rate their confidence.

Participants

Thirty-two individuals (10 males) aged 18- to 22-years-old ($M = 19.63$, $SD = 1.03$) participated in this study. We chose this sample size because it completed four rounds of counterbalancing (see Stimuli section below) and because our previous study with a similar design used an N of 28 (Doss et al., 2016). All individuals were native English speakers and had normal or corrected-to-normal vision. The institutional review board at the University of Chicago approved all experimental procedures in this study. Participants were compensated for their time with class credit.

Stimuli

Three hundred sixty pictures were selected from both the International Affective Picture System (Lang, Bradley, & Cuthbert, 2008) and the Nencki Affective Picture System (Marchewka, Żurawski, Jednoróg, & Grabowska, 2014) in an effort to balance the semantic relatedness between valence conditions. There were 120 pictures in each valence condition, and these pictures had the following mean (SD) normed valences and arousals, respectively: negative 3.18 (.53) and 5.87 (.50), neutral 5.15 (.48) and 3.71 (.88), positive 6.92 (.54) and 5.47 (.41). Negative pictures were more arousing than neutral (95% CI = [1.978, 2.342], $t(238) = 23.412$, $p < .001$, $d = 3.023$) and positive (95% CI = [.277, .509], $t(238) = 6.692$, $p < .001$, $d = .864$) pictures, and positive pictures were more arousing than neutral pictures (95% CI = [1.592, 1.942], $t(238) = 19.911$, $p < .001$, $d = 2.571$). Each picture was ascribed to a three- to five-word label to describe the content of the picture (e.g., “orangutan hanging from electrical wire”). See Supplemental Materials (SM) showing that the arousal of these labels, as well as their corresponding pictures, were rated in accordance with the normed arousal ratings of the pictures (i.e., negative > positive > neutral). In order to create the stimuli that were presented during the perceptual misinformation phase, the original 360 pictures were scrambled by segmenting each one into quadrants and then segmenting these quadrants further by subdividing them into smaller quadrants. Each smaller quadrant was then rotated in a counterclockwise fashion within the larger quadrants (Figure 1). This method for scrambling the pictures allows key features described by the label to be discerned but also prevents the participant from mistaking the scrambled version of the picture for the intact version.

The pictures were divided into 8 lists of 45 pictures (15 of each valence) based on conceptual similarities (e.g., food, animals, vehicles). The lists were rotated through the experimental conditions in a Latin square design such that for each participant there were 135 targets (45 of each valence), 135 lures (45 of each valence), and 90 nonstudied foils (30 of each valence). For both targets and lures, 90 of the 135 items (30 of each valence) were presented in the perceptual misinformation phase, and the other 45 items (15 of each valence) were presented in the conceptual misinformation phase (see below).

Procedure

The experiment consisted of four phases: an incidental encoding phase, two misinformation phases (perceptual and conceptual), and a cued recollection phase separated by a 24-hr delay from the preceding three phases. At the beginning of each phase, a researcher read the

instructions to participants, and participants also read the instructions on the computer screen.

During the encoding phase, participants were presented with 270 labels in black font on a grey background in random order. Each label was presented for 2,000 ms during which participants were instructed to rate its emotional intensity (i.e., arousal) on a 5-point scale. Emotional intensity was defined as the strength of the emotional reaction regardless of negative or positive content. For half the trials (targets), a picture was presented after the label for 2,000 ms during which participants were again to make an emotional intensity rating. For the other half of trials (lures), no picture was presented. Across all phases, each trial ended with a fixation cross for a random duration that was on average 2,000 ms and never shorter than 500 ms.

During the perceptual misinformation phase, participants were presented with 180 scrambled picture and label pairings in random order. Each trial consisted of the simultaneous presentation of a scrambled picture and its corresponding label near the bottom of the screen for 2,000 ms. Participants were instructed to mentally unscramble each picture described by the label and rate the clarity of their mental image on a five-point scale. Half of the scrambled picture and label pairings corresponded to target trials and the other half corresponded to lure trials. Labels were the same as those presented during the encoding phase.

During the conceptual misinformation phase, participants were presented with 90 labels from the encoding phase in random order. Each label was presented for 2,000 ms during which participants were instructed to rate the emotional intensity. Each label was presented twice to increase familiarity of the conceptual misinformation trials relative to the perceptual misinformation (which was only presented once), thereby providing a high-familiarity baseline for the perceptual misinformation trials. Half of the labels corresponded to target trials and the other half to lure trials. After this phase, participants left the testing room and returned the following day for the cued recollection phase.

During the retrieval phase, participants' picture memory was tested by cueing them with 270 familiar labels (135 targets and 135 lures) and 90 new labels (nonstudied foils). Participants were asked to remember whether they saw the intact version of the picture described by the label ("yes"/"no"). Participants were told that recollection of a scrambled picture was irrelevant to their recollection of an intact picture because it was possible that both an intact and a scrambled picture were presented or just a scrambled picture was presented. This procedure was intended to minimize the use of a recall-to-reject strategy. After each memory decision, participants were then asked to rate their confidence on a five-point scale. This phase was self-paced.

Code and Data Availability

Raw data and MATLAB code for scrambling pictures can be found at <https://osf.io/az7wp/> (Doss, 2018).

Experiment 1 Results and Discussion

Full data for Experiment 1 can be found in Table 1. In the following results sections, our primary analyses are on recollection errors (i.e., false alarms on the recollection task, which we assume are indicative of false recollection of nonstudied pictures, although familiarity-based guessing also may contribute). We show perceptual and conceptual misinformation impacted recollection errors across all emotional categories, demonstrating the two key effects of Doss et al. (2016) and extending these mechanisms to emotional recollection. We also show reduced recollection errors for negative lures compared to positive and neutral lures, consistent with an emotional distinctiveness heuristic. There was some evidence that this heuristic reduced the impact of the conceptual fluency for negative items, although this effect was stronger in Experiment 2. Finally, we also report more traditional measures of target-lure discrimination to objectively measure recollection accuracy, and analyses of those results are consistent with the analyses of recollection errors.

Perceptual Recombination Analysis

To measure the potential impact of a perceptual recombination mechanism, we directly compared false alarms to the two kinds of studied lures: those that had been studied in the perceptual misinformation list (thereby boosting the activation of perceptual and conceptual information) and those that had been studied in the conceptual misinformation list (thereby boosting the activation of conceptual information alone). Greater false alarms to lures from the perceptual misinformation list are taken as evidence for a perceptual recombination process while controlling for the possible influence of conceptual information, which is assessed using the conceptual misinformation lures as a baseline.

A 3 (valence: negative, neutral, positive) $\times$ 2 (misinformation: perceptual, conceptual) repeated measures ANOVA revealed a main effect of misinformation on false alarms ($F(1, 31) = 30.369, p < .001, \eta_p^2 = .495$). As is seen in Figure 2a, false alarms were greater for perceptual misinformation lures compared to the conceptual misinformation baseline. This misinformation effect was robust and did not interact with emotional valence ($F(1.390, 43.103) = .198, p > .250$; a Greenhouse-Geisser correction was applied here and for all subsequent violations of sphericity), demonstrating a perceptual recombination effect across all emotional categories. There also was a main effect of valence ($F(2, 62) = 5.877, p = .005, \eta_p^2 = .159$), as false alarms were lower for negative compared to neutral items and positive items. The reduction in false alarms for negative lures is consistent with the operation of an emotional distinctiveness heuristic: participants expected negative pictures to be recollected especially well, helping them to avoid false alarms to negative lures associated with either perceptual or conceptual misinformation. However, because this valence effect did not interact with the misinformation manipulation, there was no evidence that an emotional distinctiveness heuristic suppressed the operation of perceptual recombination.

Conceptual Fluency Analysis

To measure the conceptual fluency mechanism, false alarms to lures studied with conceptual misinformation were compared to nonstudied foils that did not have such conceptual activation (i.e., a no misinformation baseline). A 3 (valence: negative, neutral, positive) $\times$ 2

(misinformation: conceptual, no misinformation) repeated measures ANOVA revealed a main effect of misinformation ($F(1, 31) = 19.174, p < .001, \eta_p^2 = .382$). As is seen in Figure 2a, activating conceptual misinformation increased false alarms compared to the nonstudied foil baseline. There also was a main effect of valence ($F(2, 62) = 9.699, p < .001, \eta_p^2 = .238$), as false alarms for negative lures were overall lower than those for neutral and positive lures, and there was a marginal valence by misinformation interaction ($F(2, 62) = 2.736, p = .073, \eta_p^2 = .081$). This interaction was explained by fewer false alarms for negative compared to neutral misinformation lures (95% CI [.013, .078], $t(31) = 2.871, p = .007, d = .508$), with no corresponding difference in false alarms to negative and neutral nonstudied foils with no misinformation (95% CI [-.016, .003], $t(31) = 1.359, p = .184$), which were near floor. This pattern of effects for negative emotion suggests that an emotional distinctiveness heuristic may be especially useful in suppressing the influence of conceptual fluency on false recollection.

High Confidence Responses

We also considered high confidence false alarms to studied lures (i.e., the top two confidence bins). As can be seen in Figure 3, such high confidence false alarms almost never occurred for nonstudied foils, though they occurred for about 2 to 3% of lures with conceptual misinformation and 9 to 11% of lures with perceptual misinformation. A 3 (valence: negative, neutral, positive) $\times$ 2 (misinformation: perceptual, conceptual) repeated measures ANOVA revealed a main effect of misinformation type on false alarm rates ($F(1, 31) = 15.100, p < .001, \eta_p^2 = .328$, owing to perceptual misinformation eliciting greater recollection errors than conceptual misinformation. Neither the main effect of valence ($F(2, 62) = .723, p > .250$) nor the valence by misinformation interaction were significant ($F(2, 62) = .087, p > .250$), demonstrating a perceptual recombination effect for all emotional categories. A similar high confidence analysis of conceptual fluency was not conducted because of fewer high confidence false alarms to nonstudied foils, though inspection of Figure 3 indicates that this mechanism also increased high confidence errors relative to baseline for each emotional category.

Recollection Accuracy Analysis

In addition to our primary analysis of false recollection effects, we also calculated target-lure discrimination scores as a more traditional measure of recollection accuracy. Within each emotional and misinformation condition, recollection accuracy was calculated as the hit rates for targets minus the false alarm rates for studied lures (i.e., $p(\text{"yes"}|\text{target}) - p(\text{"yes"}|\text{lure})$). Because each of these targets and lures were initially studied in two different lists (i.e., the initial label/picture study list and one of the subsequent misinformation lists), they should have been highly familiar so that the accurate discrimination between the targets and studied lures should have relied primarily on recollection of the target picture.

Analysis of these recollection accuracy scores yielded effects that were consistent with those which we have already presented and discussed. A 3 (valence: negative, neutral, positive) $\times$ 2 (misinformation: perceptual, conceptual) repeated measures ANOVA on memory accuracy

revealed a main effect of valence on memory accuracy ($F(2, 62) = 13.524, p < .001, \eta_p^2 = .304$).

As can be seen in Figure 2b, there was greater memory accuracy for negative compared to neutral and positive items. This finding demonstrates an emotional superiority effect in recollection observed especially for negative material (Bowen, Kark, & Kensinger, 2018), and here it was at least partly driven by reduced recollection errors to negative lures. There also was a main effect of misinformation ($F(1, 31) = 19.276, p < .001, \eta_p^2 = .383$), as participants were more accurate at differentiating targets and lures associated with conceptual compared to perceptual misinformation. This effect suggests that perceptual misinformation manipulation was more effective at generating false recollections than the conceptual fluency manipulation, at least under the conditions of this experiment. Finally, the valence by misinformation interaction was not significant on recollection accuracy scores ($F(2, 62) = .637, p > .250$), although as discussed we did find a marginal interaction with our conceptual fluency analysis, and we revisit this effect in Experiment 2.

Experiment 2

The goal of Experiment 2 was to investigate the impact of perceptual and conceptual misinformation when the misinformation occurred on the same day as retrieval instead of immediately after the encoding phase as in Experiment 1. This experiment was identical to Experiment 1 except the 24-hour delay was introduced before the misinformation phases rather than before the retrieval phase. Doss et al. (2016) found that the effect of conceptual misinformation on recollection errors was larger when conceptual activation occurred just before the test phase as opposed to the day before (cf. Doss, Picart, & Gallo, 2018 in which activating conceptual associations, via context reinstatement, prior to memory retrieval at the trial level robustly increased high confidence false memories). Because perceptual misinformation was not affected by such a timing manipulation, these findings suggest that conceptual fluency fades more rapidly over time. This experiment, therefore, provided a replication attempt of Experiment 1 under conditions where conceptual misinformation effects might be even greater.

Participants

Thirty-two individuals (18 males) aged 18- to 22-years-old ($M = 19.84, SD = 1.13$) were recruited from the University of Chicago and surrounding community with the same inclusion criteria as Experiment 1.

Stimuli

The stimuli were identical to those used in Experiment 1.

Procedure

The experiment was identical to Experiment 1 except the 24-hour delay was introduced immediately after the encoding phase so that the misinformation phases always occurred on the second day (i.e., just prior to the retrieval phase).

Experiment 2 Results and Discussion

Full data for Experiment 2 can be found in Table 2. Our results sections follow the same format as Experiment 1. To anticipate these results, Experiment 2 replicated all of the key effects from Experiment 1 and also yielded stronger evidence that emotion can suppress the impact of conceptual fluency on recollection errors.

Perceptual Recombination Analysis

To investigate the impact of perceptual misinformation, we conducted a 3 (valence: negative, neutral, positive) $\times$ 2 (misinformation: perceptual, conceptual) repeated measures ANOVA (Figure 2a). There was a main effect of misinformation ($F(1, 31) = 44.526, p < .001, \eta_p^2 = .590$), as perceptual misinformation increased false alarms above the conceptual misinformation baseline. Second, this effect did not interact with valence ($F(2, 62) = 1.793, p = .175$), again demonstrating a robust perceptual recombination effect for all emotional categories. Finally, there was a main effect of valence ($F(2, 62) = 4.571, p = .014, \eta_p^2 = .129$), owing to fewer false alarms to negative compared to neutral and positive items. As in Experiment 1, this effect is consistent with an emotional distinctiveness heuristic.

Conceptual Fluency Analysis

To investigate conceptual fluency, we conducted a 3 (valence: negative, neutral, positive) $\times$ 2 (misinformation: conceptual, no misinformation) repeated measures ANOVA. This analysis revealed a main effect of misinformation ($F(1, 32) = 24.688, p < .001, \eta_p^2 = .443$). As is shown in Figure 2a, conceptual misinformation increased false alarms above the nonstudied baseline, implicating a conceptual fluency mechanism. There also was a main effect of valence ($F(2, 62) = 3.816, p = .027, \eta_p^2 = .110$), as false alarms were lowest for negative items. There was also a significant interaction between misinformation type and valence ($F(2, 62) = 3.340, p = .042, \eta_p^2 = .097$), consistent with the similar (albeit marginal) pattern observed in Experiment 1. There were fewer false alarms for negative compared to neutral misinformation items (95% CI = [.019, .127], $t(31) = 2.752, p = .010, d = .486$) with no corresponding difference between negative and neutral nonstudied foils with no misinformation, which were near floor (95% CI = [-.027, .005], $t(31) = 1.457, p = .155$). A similar pattern of findings was observed when comparing positive and neutral false alarms (conceptual misinformation: 95% CI = [-.005, .122], $t(31) = 1.877, p = .070, d = .332$; no misinformation: 95% CI = [-.011, .044], $t(31) = 1.245, p = .223, d = .220$). These effects of negative and positive emotion suggest that, when the misinformation occurred just prior to the test, an emotional distinctiveness heuristic reduced the conceptual fluency effect on recollection errors.

High Confidence Responses

A 3 (valence: negative, neutral, positive) $\times$ 2 (misinformation: perceptual, conceptual) repeated measures ANOVA on high confidence false alarms revealed a main effect of misinformation ($F(1, 31) = 32.262, p < .001, \eta_p^2 = .510$), as perceptual misinformation increased high confidence false alarms relative to conceptual misinformation (Figure 3).

There was no significant effect of valence ($F(2, 62) = .885, p > .250$) and no significant interaction ($F(1.643, 50.942) = .175, p > .250$), demonstrating a robust effect of perceptual recombination for all emotional categories. Similar to Experiment 1, inspection of Figure 3 reveals elevated high confidence false alarms as a result of conceptual misinformation, though we again did not analyze these data due to the floor levels of high confidence false alarms to nonstudied foils.

Recollection Accuracy Analysis

A 3 (valence: negative, neutral, positive) $\times$ 2 (misinformation: perceptual, conceptual) repeated measures ANOVA on recollection accuracy scores revealed a main effect of valence ($F(2, 62) = 22.619, p < .001, \eta_p^2 = .422$). As in Experiment 1, recollection accuracy was greater for negative than neutral and positive items (Figure 2b). Unlike Experiment 1, there was no significant effect of misinformation on recollection accuracy ($F(1, 31) = .023, p > .250$), potentially indicating that the two manipulations were equally likely to cause recollection errors under the conditions of this experiment. The valence by misinformation interaction failed to reach significance ($F(2, 62) = 2.608, p = .082, \eta_p^2 = .078$), although differences in effect sizes suggest that the superiority of recollection accuracy for negative material was greater for items with conceptual (negative vs. neutral: 95% CI = [.140, .290], $t(31) = 5.839, p < .001, d = 1.032$) compared to perceptual (negative vs. neutral: 95% CI = [.033, .163], $t(31) = 3.088, p = .004, d = .546$) misinformation. This pattern is consistent with the idea that negative recollections were less impacted by the corrupting influence of conceptual fluency than neutral recollections.

Between-Experiments Comparisons

In Experiments 1 and 2 we found evidence that perceptual recombination and conceptual fluency increased recollection errors across all emotional categories. Although the perceptual recombination effect was robust for all emotional categories in both experiments, there was evidence in both experiments that the conceptual misinformation effect was reduced for emotional items, especially negative items, relative to neutral items. The interaction associated with this effect was marginal in Experiment 1 and significant in Experiment 2. To directly test differences between the timing of misinformation and increase power in detecting emotional interactions, we did a combined analysis of Experiments 1 and 2.

Perceptual Recombination Analysis

A 3 (valence: negative, neutral, positive) $\times$ 2 (misinformation: perceptual, conceptual) $\times$ 2 (experiment: pre-delay misinformation, post-delay misinformation) mixed ANOVA on false alarms revealed a main effect of misinformation type ($F(1, 62) = 71.908, p < .001, \eta_p^2 = .537$), as perceptual misinformation increased recollection errors above the conceptual misinformation baseline (Figure 2a). There also was a main effect of valence ($F(2, 124) = 8.924, p < .001, \eta_p^2 = .126$) because false alarms for negative items were lower than neutral and positive items. The main effect of experiment was not significant ($F(1, 62) = 2.764, p = .101$), and none of the two- or three-way interactions were significant (all F s < 1.250 , all

$ps > .250$), demonstrating that the perceptual recombination effect was not dependent on the timing of the misinformation, and also demonstrating that the perceptual recombination effect was of similar magnitude for each emotional condition. Finally, this combined analysis also yielded a similar pattern of effects in high confidence responses, including a robust perceptual recombination effect (Figure 3; $F(1, 62) = 40.543$, $p < .001$, $\eta_p^2 = .395$).

Conceptual Fluency Analysis

A 3 (valence: negative, neutral, positive) $\times$ 2 (misinformation: conceptual, no misinformation) $\times$ 2 (experiment: pre-delay misinformation, post-delay misinformation) mixed ANOVA on false alarms revealed a main effect of misinformation condition ($F(1, 62) = 43.288$, $p < .001$, $\eta_p^2 = .411$), as conceptual misinformation increased false alarms relative to the nonstudied foil baseline. There also was a main effect of experiment ($F(1, 62) = 5.606$, $p = .021$, $\eta_p^2 = .083$), as false alarms were overall greater in Experiment 2. Unlike Doss et al. (2016), the size of the conceptual misinformation effect did not interact with the timing of misinformation ($F(1, 62) = 2.493$, $p = .119$), although inspection of Figure 2a suggests that for neutral items (as in Doss et al.) this effect was larger in Experiment 2 than in Experiment 1. The main effect of valence was also significant ($F(1, 32) = 8.849$, $p < .001$, $\eta_p^2 = .125$), as false alarms for negative items were lower than those for neutral items or positive items. Finally, there was a significant interaction between misinformation and valence ($F(2, 124) = 4.283$, $p = .016$, $\eta_p^2 = .065$), but the timing of misinformation did not interact with valence ($F(2, 124) = 1.730$, $p = .181$) nor was the three-way interaction significant ($F(2, 124) = 2.071$, $p = .130$). The interaction between misinformation and valence indicates that false alarms were greater for neutral compared to negative items with conceptual misinformation (95% CI = [.029, .090], $t(63) = 3.846$, $p < .001$, $d = .481$), but this effect was only marginal for items with no misinformation (95% CI = [-.000, .018], $t(63) = 1.954$, $p = .055$, $d = .244$). By contrast, neither of these effects reached significance when comparing positive and neutral items (conceptual misinformation: 95% CI = [-.013, .060], $t(63) = 1.312$, $p = .194$; no misinformation: 95% CI = [-.001, .031], $t(63) = 1.915$, $p = .060$, $d = .239$). These results thus provide additional evidence that the conceptual fluency effect was reduced for negative items relative to neutral items. A high confidence analysis testing the conceptual fluency mechanism was not conducted, as high confidence false alarms for nonstudied foils were at floor.

General Discussion

This study investigated the impact of two hypothetical mechanisms of memory construction—perceptual recombination and conceptual fluency—on the accuracy of neutral and emotional recollections. Here we highlight three key findings from each of the two experiments. First, we found evidence that the two mechanisms increased recollection errors, using different pictorial materials and other modifications of the Doss et al. (2016) procedure, thereby replicating those effects and demonstrating the robust nature of these two mechanisms across the emotional categories. Perceptual recombination, in particular, was found to drive high confidence errors, implicating a role in creating vivid false recollections. Second, in each experiment, we found that the size of the perceptual misinformation effect

on recollection errors was of similar magnitude across the emotional categories. Third, we found that the size of the conceptual fluency effect was reduced for negative (and to some degree positive) compared to neutral items, especially when the conceptual fluency manipulation occurred just prior to memory retrieval. This finding was consistent with the use of an emotional distinctiveness heuristic, as was the overall lower recollection error rate for negative compared to neutral lures. Below we elaborate on the theoretical significance of each of these findings.

Independent from our emotional manipulations, our experiments provide a conceptual replication of several key findings from Doss et al. (2016). False alarms were elevated for lures that had been associated with perceptual misinformation compared to the conceptual fluency baseline, suggesting a perceptual recombination effect on false recollection. Moreover, false alarms were elevated for lures that had been associated with conceptual fluency relative to the nonstudied foil baseline, suggesting a conceptual fluency effect on false recollection. Similar to Doss et al., we also found that the perceptual misinformation effect was not affected by the timing of the misinformation (i.e., pre- or post-delay), although evidence was mixed with respect to the extent that conceptual misinformation had a larger effect when administered post-delay compared to pre-delay. In addition to extending these effects to a different set of materials, the current experiments demonstrate that mental imagery of pictures during the initial study phase is unnecessary to obtain these effects, as we eliminated this aspect of the Doss et al. procedure.

With respect to the impact of emotion on these effects, we found that the perceptual misinformation effect was a similar magnitude across all emotional categories, but the size of the conceptual misinformation effect varied with emotionality. In Experiment 2, we found a reduced conceptual misinformation effect for both negative and positive items, and the combined analysis found a reduced effect for negative items. This effect implicates a reduced conceptual fluency effect for emotional items compared to neutral items, as predicted by the distinctiveness heuristic hypothesis and consistent with some of the results of Choi et al (2013). The lack of a similar reduction of the perceptual misinformation effect suggests that this effect may benefit less from a distinctiveness heuristic than the conceptual fluency effect, potentially because the presentation of fragmented pictures provides more perceptually detailed false recollections that mitigates the logic of a distinctiveness heuristic (i.e., “I would remember a perceptually detailed picture if it had been studied.”). By contrast, there was no evidence for the emotional enhancement hypothesis or the idea that emotional misinformation might be better remembered than neutral misinformation and hence have a larger effect on either of the two constructive mechanisms.

Related to the idea of an emotional distinctiveness heuristic, in each experiment we also found an overall tendency for fewer false alarms to negative lures, including those that had been associated with perceptual misinformation, conceptual misinformation, or no misinformation (nonstudied foils). Each of these effects is consistent with reduced impact of concept fluency or familiarity on negative false recollections, potentially owing to a distinctiveness heuristic because all lures could have been somewhat affected by concept fluency including nonstudied foils that could be familiar from idiosyncratic conceptual or semantic associations across items. Positive items did not consistently show this effect,

potentially because these items were less arousing than negative items and/or because they were more easily envisioned, as demonstrated in the analysis of clarity ratings for scrambled pictures from the perceptual misinformation phase (see SM). The overall impact of a distinctiveness heuristic may also have been greater for negative items because these items elicited stronger recollections as demonstrated by a concurrent increase in hit rates and recollection accuracy.

To summarize, we found evidence that both perceptual recombination and conceptual fluency can drive recollection errors of neutral and emotional events. We also found evidence that an emotional distinctiveness heuristic can mitigate the impact of conceptual fluency on recollection errors, but there was no evidence that an emotional distinctiveness heuristic affected perceptual recombination. Outside of the lab, these results suggest that detailed false recollections of negative events are more difficult to implant based on hearsay or misleading statements alone, compared to the presentation of perceptual misinformation, though certainly not impossible, as the effects of conceptual fluency were still quite apparent regardless of how a memory is emotionally charged. These results indicate that emotion can differentially impact the different constructive mechanisms that drive episodic recollection, and they help to explain why emotion sometimes enhances and other times diminishes memory distortion.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Figure 1.
Examples of both the intact and scrambled version of pictures from the stimulus set.

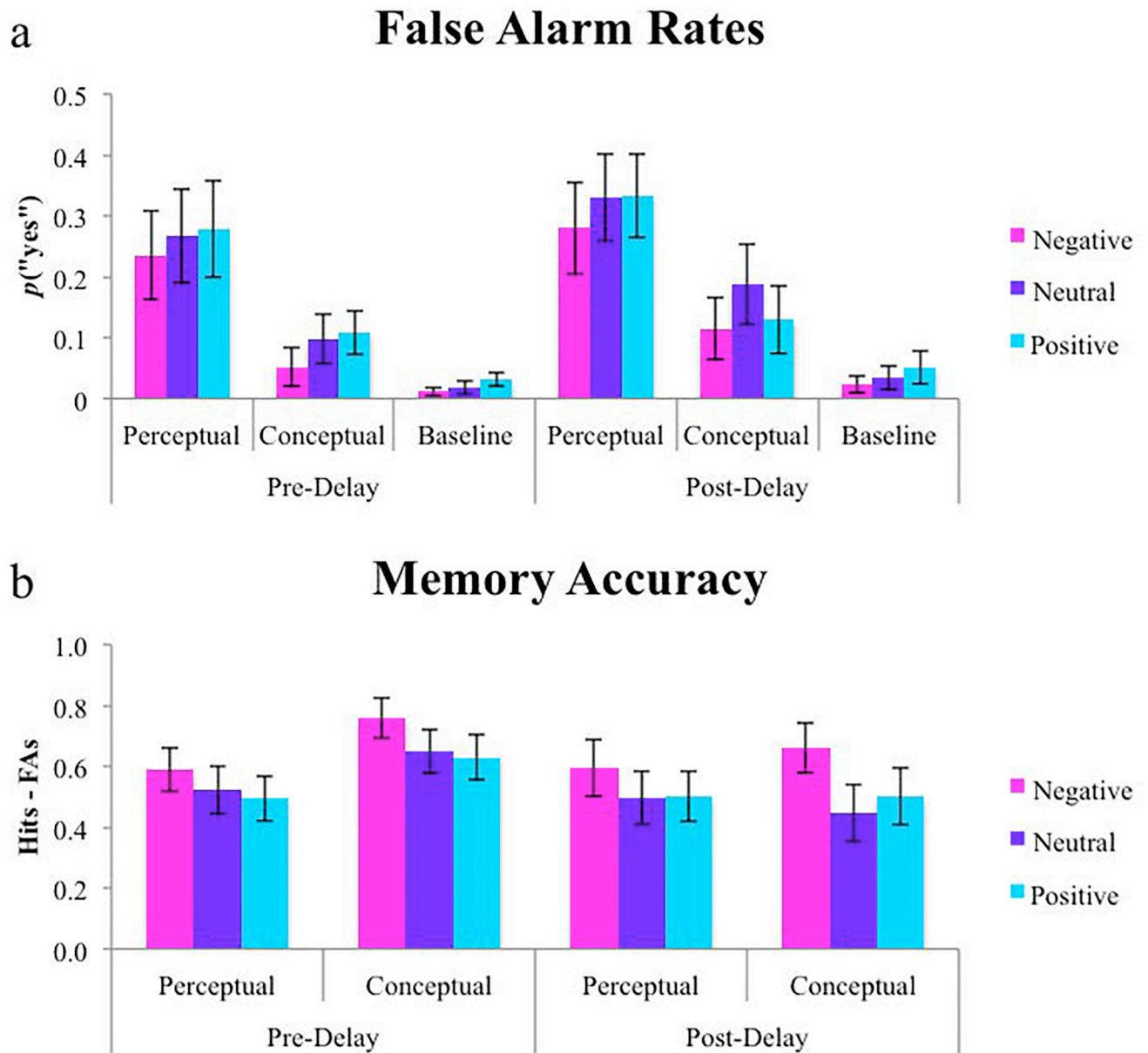


Figure 2. False alarm rates (a) and memory accuracy (b) on the cued recollection task for each emotional valence with pre-delay (Experiment 1) and post-delay (Experiment 2) misinformation.

High Confidence False Alarm Rates

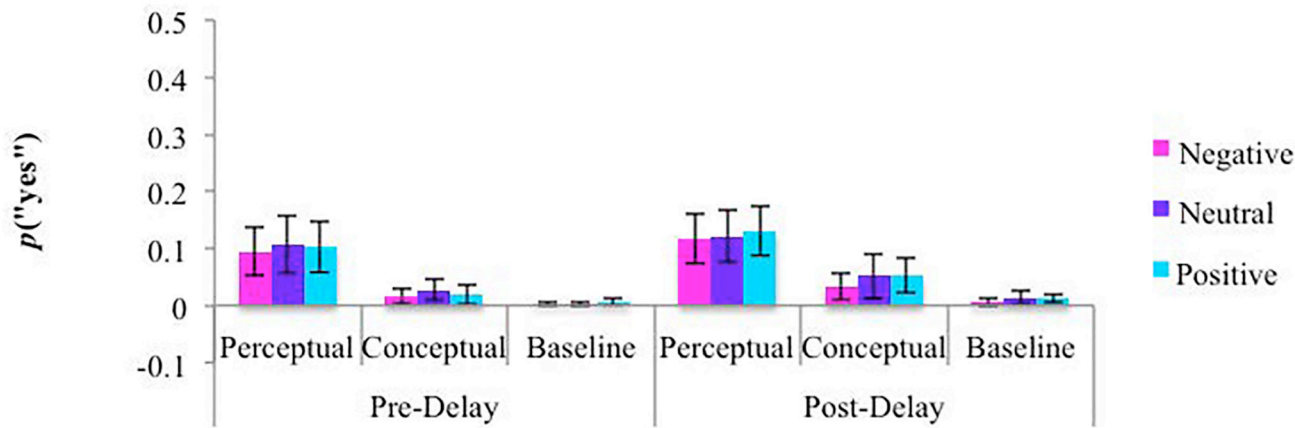


Figure 3. High confidence (top two levels of confidence) false alarm rates on the cued recollection task for each emotional valence with pre-delay (Experiment 1) and post-delay (Experiment 2) misinformation.

Table 1.

Average arousal ratings for labels and pictures during encoding phase, average clarity ratings for scrambled pictures during perceptual misinformation phase, average arousal ratings for labels during conceptual misinformation phase, hit rates, false alarm rates, and memory accuracy (hit rate - false alarm rates) for Experiment 1 (i.e., pre-delay misinformation). Mean [95% confidence interval].

	Perceptual			Conceptual			No Misinformation		
	Negative	Neutral	Positive	Negative	Neutral	Positive	Negative	Neutral	Positive
Enc Lab Aro Rating (targets)	2.93 [2.63, 3.23]	1.78 [1.59, 1.98]	2.15 [1.90, 2.40]	2.90 [2.59, 3.22]	1.91 [1.68, 2.14]	2.21 [1.95, 2.48]	-	-	-
Enc Lab Aro Rating (lures)	2.82 [2.55, 3.09]	1.83 [1.63, 2.03]	2.18 [1.94, 2.43]	2.83 [2.52, 3.14]	1.84 [1.59, 2.08]	2.22 [1.92, 2.52]	-	-	-
Enc Pic Aro Rating (targets)	3.50 [3.23, 3.77]	1.96 [1.72, 2.19]	2.74 [2.46, 3.01]	3.71 [3.42, 4.00]	2.06 [1.79, 2.32]	2.71 [2.39, 3.03]	-	-	-
Mis Rating (targets)	3.46 [3.25, 3.68]	3.42 [3.22, 3.61]	3.67 [3.47, 3.86]	3.31 [3.01, 3.61]	1.87 [1.63, 2.11]	2.42 [2.12, 2.72]	-	-	-
Mis Rating (lures)	2.61 [2.40, 2.82]	2.78 [2.58, 2.98]	2.99 [2.79, 3.19]	3.01 [2.72, 3.31]	1.78 [1.56, 1.99]	2.27 [1.99, 2.55]	-	-	-
Hit Rate	.82 [.78, .86]	.79 [.74, .84]	.77 [.72, .82]	.81 [.75, .87]	.75 [.69, .81]	.74 [.67, .80]	-	-	-
FA Rate	.24 [.16, .31]	.27 [.19, .34]	.28 [.20, .36]	.05 [.02, .08]	.10 [.06, .14]	.11 [.07, .14]	.01 [.00, .02]	.02 [.01, .03]	.03 [.02, .04]
Accuracy	.59 [.52, .66]	.52 [.44, .60]	.49 [.42, .57]	.76 [.69, .83]	.65 [.58, .72]	.63 [.55, .70]	-	-	-
Hi Conf Hit Rate	.65 [.59, .71]	.57 [.50, .64]	.59 [.52, .65]	.67 [.61, .74]	.55 [.48, .62]	.55 [.48, .63]	-	-	-
Hi Conf FA Rate	.09 [.05, .14]	.11 [.06, .16]	.10 [.06, .15]	.02 [.00, .03]	.03 [.01, .05]	.02 [.00, .04]	.00 [.00, .01]	.00 [.00, .01]	.01 [.00, .01]
Hi Conf Accuracy	.56 [.50, .62]	.46 [.39, .53]	.48 [.42, .55]	.66 [.59, .72]	.53 [.45, .60]	.54 [.46, .61]	-	-	-

Enc = encoding, Lab = label, Aro = arousal, Mis = misinformation, FA = false alarm, Hi Conf = high confidence.

Table 2.

Average arousal ratings for labels and pictures during encoding phase, average clarity ratings for scrambled pictures during perceptual misinformation phase, average arousal ratings for labels during conceptual misinformation phase, hit rates, false alarm rates, and memory accuracy (hit rate - false alarm rates) for Experiment 2 (i.e., post-delay misinformation). Mean [95% confidence interval].

	Perceptual			Conceptual			No Misinformation		
	Negative	Neutral	Positive	Negative	Neutral	Positive	Negative	Neutral	Positive
Enc Lab Aro Rating (targets)	2.96 [2.74, 3.18]	1.95 [1.76, 2.13]	2.36 [2.16, 2.57]	2.90 [2.65, 3.15]	1.90 [1.69, 2.11]	2.36 [2.06, 2.65]	-	-	-
Enc Lab Aro Rating (lures)	3.02 [2.79, 3.25]	1.91 [1.72, 2.09]	2.28 [2.10, 2.47]	2.96 [2.68, 3.23]	2.00 [1.80, 2.20]	2.44 [2.18, 2.69]	-	-	-
Enc Pic Aro Rating (targets)	3.62 [3.37, 3.88]	2.20 [1.99, 2.41]	2.79 [2.54, 3.04]	3.58 [3.26, 3.89]	2.11 [1.87, 2.35]	2.87 [2.53, 3.21]	-	-	-
Mis Rating (targets)	3.03 [2.81, 3.24]	3.11 [2.90, 3.32]	3.26 [3.06, 3.46]	3.14 [2.87, 3.41]	1.98 [1.74, 2.21]	2.45 [2.12, 2.79]	-	-	-
Mis Rating (lures)	2.49 [2.26, 2.71]	2.72 [2.50, 2.95]	2.86 [2.61, 3.11]	3.19 [2.94, 3.45]	2.04 [1.79, 2.28]	2.60 [2.29, 2.91]	-	-	-
Hit Rate	.87 [.84, .91]	.83 [.77, .88]	.83 [.79, .88]	.78 [.72, .83]	.63 [.56, .71]	.63 [.55, .71]	-	-	-
FA Rate	.28 [.20, .36]	.33 [.26, .40]	.33 [.26, .40]	.11 [.06, .17]	.19 [.12, .25]	.13 [.07, .18]	.02 [.01, .04]	.03 [.01, .05]	.05 [.02, .08]
Accuracy	.59 [.50, .69]	.50 [.41, .58]	.50 [.42, .58]	.66 [.58, .74]	.45 [.35, .54]	.50 [.41, .59]	-	-	-
Hi Conf Hit Rate	.77 [.71, .83]	.67 [.60, .75]	.66 [.59, .73]	.64 [.56, .71]	.45 [.36, .54]	.44 [.35, .53]	-	-	-
Hi Conf FA Rate	.12 [.07, .16]	.12 [.08, .17]	.13 [.09, .17]	.03 [.01, .06]	.05 [.01, .09]	.05 [.02, .08]	.01 [.00, .01]	.01 [.00, .03]	.01 [.00, .02]
Hi Conf Accuracy	.65 [.57, .72]	.55 [.47, .63]	.53 [.45, .60]	.60 [.52, .68]	.40 [.30, .49]	.39 [.30, .48]	-	-	-

Enc = encoding, Lab = label, Aro = arousal, Mis = misinformation, FA = false alarm, Hi Conf = high confidence.